THERMOPAC ROI ANALYSIS REPORT

Generated on 6/25/2025

PROJECT SUMMARY

Customer: ENDA UK

Project: ENDA UOR 6075

Plant Capacity: 6000 LPH

Currency: GBP

Annual Processing: 52,560,000 Liters

FINANCIAL SUMMARY

Investment Breakdown

Plant Cost: GBP 6,940,000

Tank Farm: GBP 1,293,800

Additional Equipment: GBP 1,056,000

Other Costs: GBP 3,481,000

Total Investment: GBP 12,770,800

Annual Revenue Analysis

Light Base Oil: 50% = 26280 tons × GBP 444 = GBP 11,668,320

Heavy Base Oil: 22% = 11563 tons × GBP 466 = GBP 5,388,451.2

Naphtha/Gas Oil: 7% = 3679 tons × GBP 296 = GBP 1,089,043.2

Residue: 15% = 7884 tons × GBP 222 = GBP 1,750,248

Waste Water: 5% = 2628 tons × GBP -40 = GBP -105,120

Total Annual Revenue: GBP 19,790,942.4

Annual Operating Costs

Feedstock: GBP 720,000/month × 12 = GBP 8,640,000

Power: GBP 15,120/month × 12 = GBP 181,440

Fuel: GBP 108,000/month × 12 = GBP 1,296,000

Consumables: GBP 18,000/month × 12 = GBP 216,000

Labor: GBP 30,000/month × 12 = GBP 360,000

Maintenance: GBP 13,707/month × 12 = GBP 164,484

Total Operating Costs: GBP 10,857,924

KEY PERFORMANCE INDICATORS

Annual Profit: GBP 8,933,018.4

Payback Period: 1.4 years

Annual ROI: 69.9%

Profit Margin: 45.1%

DETAILED USER INPUTS BY STEP

STEP 1: PLANT CONFIGURATION

Customer Name: ENDA UK
Project Name: ENDA UOR 6075
Plant Capacity: 6000 LPH
Currency: GBP
Plant Operation Days (Monthly): 25 days
Working Capital Requirement: GBP 0
Project Cost (Local): GBP 5,482,600
Exchange Rate: N/A
Interest Rate on Debt: N/A% annual

Additional Project Costs:

- Freight & Insurance: GBP 55,000
- Import Duty & VAT: GBP 274,000
- Plot Cost: GBP 548,000
- Civil Cost: GBP 439,000
- Refinery Shed: GBP 548,000
- Utility Shed: GBP 439,000
- Office Building: GBP 274,000
- Mechanical & Electrical: GBP 548,000
- Legal Fees: GBP 55,000
- Pre Formation Expenses: GBP 27,000
- Commissioning & Travel: GBP 164,000
- Contingency: GBP 110,000

STEP 2: TANK FARM & UTILITIES

Tank Configuration:

- Tank 1: 3x 600KL = 0 GBP
- Tank 2: 1x 600KL = 0 GBP
- Tank 3: 1x 500KL = 0 GBP
- Tank 4: 1x 200KL = 0 GBP
- Tank 5: 1x 200KL = 0 GBP
- Tank 6: 2x 400KL = 0 GBP
- Tank 7: 1x 600KL = 0 GBP
- Tank 8: 1x 100KL = 0 GBP
- Tank 9: 1x 300KL = 0 GBP
- Tank 10: 1x 100KL = 0 GBP
- Tank 11: 1x 100KL = 0 GBP
- Tank 12: 2x 500KL = 0 GBP
- Tank 13: 1x 200KL = 0 GBP

Utilities Configuration:

- undefined: 1 undefined = 60,000 GBP
- undefined: 2 undefined = 180,000 GBP
- undefined: 1 undefined = 0 GBP

STEP 3: ADDITIONAL EQUIPMENT

Additional Pumps, Filters & Cooler: GBP 82,000
Tank Level Transmitters & Accessories: GBP 55,000
Pipes, Valves & Flanges: GBP 164,000
Electrical Cables & Accessories: GBP 110,000
PCC & MCC Panels: GBP 110,000
Chimney & Ducting: GBP 55,000
Cooling Tower: GBP 41,000
Diesel Generator: GBP 137,000
Quality Control Equipment: GBP 55,000
Thermic Fluid: GBP 110,000
Expansion & Structure: GBP 82,000
Crane Hire Charges: GBP 27,000
Labor Erection & Commissioning: GBP 28,000

STEP 4: OPERATING COSTS (MONTHLY)

Feedstock Cost per Liter: GBP 0.2
Power Cost: GBP 15,120/month
Fuel Cost: GBP 108,000/month
Consumables Cost: GBP 18,000/month
Labor Cost: GBP 30,000/month
Maintenance Cost: GBP 13,707/month

STEP 5: PRODUCT YIELDS & PRICING

Naphtha/Gas Oil: 7% yield @ GBP 296/ton
Light Base Oil: 50% yield @ GBP 444/ton
Heavy Base Oil: 22% yield @ GBP 466/ton
Residue: 15% yield @ GBP 222/ton
Waste Water: 5% yield @ GBP -40/ton
Process Loss: 1% yield @ GBP 0/ton

STEP 6: FINANCIAL CONFIGURATION

CALCULATION SUMMARY

Total Investment from All Steps: GBP 10,259,600
Monthly Operating Costs: GBP 904,827
Annual Operating Costs: GBP 10,857,924
Total Product Yields: 100.0%
Plant Utilization: 25 days/month

PROJECT COST BREAKDOWN (STEP 1)

Project Information

Project Name: ENDA UOR 6075

Customer Name: ENDA UK

Plant Capacity: 6000 LPH

Selected Currency: GBP

Base Plant Cost

Base Plant Cost (USD): 6,940,000

Base Plant Cost (GBP): 5,482,600

Additional Project Costs

Cost Component	Amount (GBP)
Freight & Insurance	55,000
Import Duty & VAT	274,000
Plot Cost	548,000
Civil Cost	439,000
Refinery Shed	548,000
Utility Shed	439,000
Office Building	274,000
Mechanical & Electrical	548,000
Fire Suppression System	274,000
Insulation Cost	110,000
Legal Fees	55,000
Pre Formation Expenses	27,000
Commissioning & Travel	164,000
Contingency	110,000

Total Additional Costs	3,865,000

Step 1 Cost Summary

Base Plant Cost: GBP 5,482,600

Additional Costs: GBP 3,865,000

Total Step 1 Investment: GBP 9,347,600

TANK FARM & UTILITIES COST BREAKDOWN

Tank Farm Details

Tank Description	% Capacity	Storage Days	Required (KL)	Tank Size (KL)	Quantity	Cost/Tank (GBP)	Total Cost (GBP)
Used oil storage tanks	150	7	1600	600	3		
Light Base Oil	80	5	600	600	1		
Heavy Base Oil	60	5	500	500	1		
Light Base Oil Intermediate tank	40	3	200	200	1		
Heavy Base Oil Intermediate tank	30	3	150	200	1		
Finish Light Base Oil	70	7	800	400	2		
Finish Heavy Base Oil	50	7	600	600	1		
Waste Water tank	25	2	100	100	1		
Naphtha / Gas Oil storage tank	35	4	250	300	1		
Residue storage tank	20	3	100	100	1		

Process water tank	30	2	100	100	1		
Fire water storage tank	20	30	900	500	2		
Fuel oil Tank	15	5	150	200	1		

Utilities & Equipment Details

Equipment	Specifications	Quantity	Unit Cost (GBP)	Total Cost (GBP)
		1		60,000
		2		180,000
		1		

Tank Farm & Utilities Summary

Total Tank Farm Cost: GBP 0

Total Utilities Cost: GBP 240,000

Combined Tank Farm & Utilities: GBP 240,000

ADDITIONAL EQUIPMENT BREAKDOWN (STEP 3)

Additional Equipment Details

Equipment Component	**Cost (GBP)**
Pipes, Valves & Flanges	164,000
Tank Level Transmitters	55,000
Additional Pumps & Filters	82,000
Quality Control Equipment	55,000
Labor Erection & Commissioning	28,000
Electrical Cables & Accessories	110,000
Total Additional Equipment	**494,000**

Step 3 Equipment Summary

Total Additional Equipment Cost: GBP 494,000

OPERATING COSTS BREAKDOWN (STEP 4)

Cost Component	**Amount**
Feedstock Cost per Liter	GBP 0.2
Power Cost (Monthly)	GBP 15,120
Fuel Cost (Monthly)	GBP 108,000
Chemical Cost (Monthly)	GBP 18,000
Labor Cost (Monthly)	GBP 30,000
Maintenance Cost (Monthly)	GBP 13,707

PRODUCT YIELDS & PRICING (STEP 5)

Product	Yield (%)	Price (GBP)
Naphtha / Gas Oil	7.0%	296
Light Base Oil	50.0%	444
Heavy Base Oil	22.0%	466
Residue	15.0%	222
Waste Water	5.0%	-40
Process Loss	1.0%	0
Total Yield	**100.0%**	

PROFIT & LOSS STATEMENT (ANNUAL)

Description	Amount (GBP)
REVENUE	**19.5M**
Naphtha / Gas Oil	1.1M
Light Base Oil	11.5M
Heavy Base Oil	5.3M
Residue	1.7M
Waste Water	-103,680

COST OF GOODS SOLD	**10.4M**
Feedstock Cost	10.4M

GROSS PROFIT	**9.2M**

OPERATING EXPENSES	**2.2M**
Power Cost	181,440
Fuel Cost	1.3M
Chemical Cost	216,000
Labor Cost	360,000
Maintenance Cost	164,484

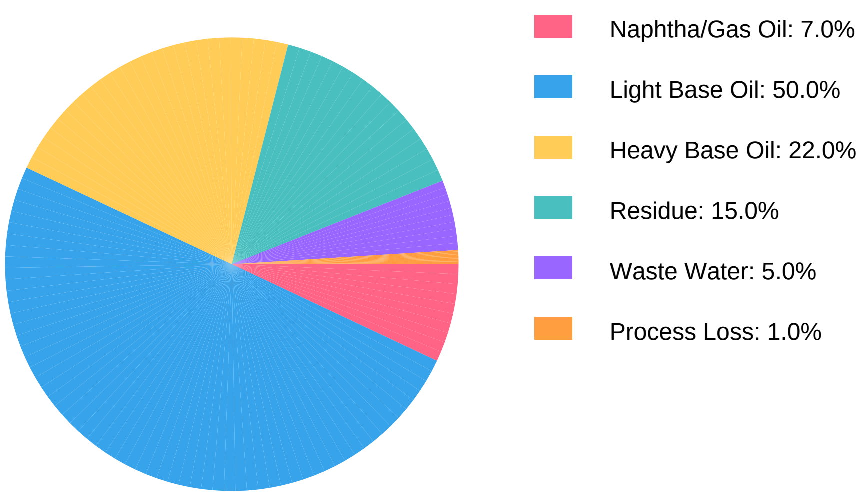
EBITDA	**6.9M**
NET PROFIT	**6.9M**

KEY FINANCIAL METRICS

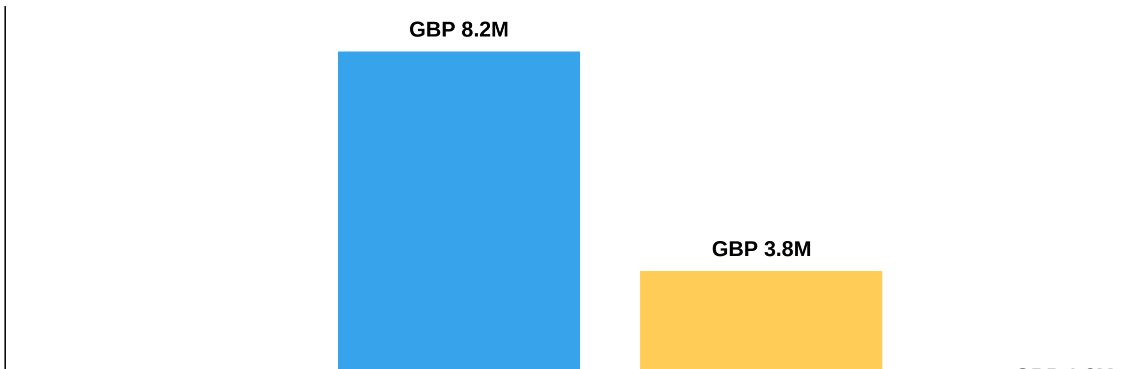
Financial Metric	Value
Gross Margin	46.88%
Net Margin	35.52%
Annual ROI	145.10%
Payback Period	0.69 years
IRR	11.18%
NPV	GBP 40,339,814

GRAPHICAL SUMMARY

Product Yield Breakdown (%)



Annual Revenue by Product (GBP)



FINANCIAL SUMMARY TABLE

Metric	Value	Unit
Plant Capacity	6,000	LPH
Operating Days	25	days/month
Annual Processing	36720	tons/year
Total CAPEX	10,259,600	GBP
Monthly OpEx	904,827	GBP/month
Annual Revenue	13,899,988.8	GBP/year
Product Yield	100.0	%
ROI	145.1	%
Payback Period	0.7	years

This comprehensive report includes all user inputs, calculations, and visual analysis from the ROI Calculator. Charts and tables reflect real-time values based on your specific project configuration.